

# Lab 8: Assumption-Lean Regression

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# Study Context and Objective

- **Program:** First Steps (Washington State, USA, 1989).
- **Goal:** Improve maternal and infant health outcomes for low-income pregnant women.
- **Specific Aim:** Reduce the incidence of low birth weight (LBW), defined as birth weight  $< 2500$  grams.
- **Study Objectives:**
  - 1 Estimate the effect of First Steps participation on the risk of LBW.
  - 2 Explore association between First Steps and maternal age.

# Data Description: King County Birth Data (2001)

## Sample:

- 2,500 singleton births
- Source: Birth certificates from King County, Washington (2001)

## Variables:

- **Child:** Gender, Singleton status, Birth weight
- **Mother:** Age, Race, Marital status, Education (years)
- **Maternal Behavior:** Smoking, Alcohol use, Pre-pregnancy weight, Weight gain
- **Pregnancy:** Parity, Gestation (weeks)
- **Programs:** Participation in First Steps and Welfare programs

# Assumption-Lean Modeling

Assume that interest lies in the association between a possibly continuous variable or exposure  $A$  and an arbitrary outcome  $Y$ , conditional on measured variables  $L$ :

$$E(Y | A, L) = \alpha(L) + \beta A,^1$$

- If we go straight fitting this model and it is **misspecified**, we are not guaranteed to capture the association between  $A$  and  $Y$ .
- **Data-adaptive** inference is not valid.

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<sup>1</sup>For a more general case, we will consider

$$g\{E(Y | A, L)\} = \alpha(L) + \beta A,$$

# Assumption-Lean Modeling - Cont'd

Consider a (semi-parametric) linear structural mean model of the form

$$E(Y^a | L) = \underbrace{\alpha(L)}_{\text{unknown}} + \underbrace{\beta(L)}_{\text{unknown}} a.$$

We then **summarize** the distribution of  $\beta(L)$  instead of assuming  $\beta(L) = \beta$ .

# Assumption-Lean Modeling - Cont'd

- 1 Define a **model free estimand**.
  - First, when the parametric assumptions hold, it should **reduce to the target parameter of interest**.
  - Second, it should be **generic**, in the sense of being well defined regardless of whether  $A$  is continuous or discrete.<sup>2</sup>
  - Third, the **estimand must capture what one is aiming for** (e.g. a conditional association), even when the model is misspecified.
- 2 Estimate it nonparametrically (based on its **efficient influence function**).

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<sup>2</sup>It should also be generic in the sense that its efficient influence function should not demand the modeling of a (conditional) density, as flexible statistical or machine learning techniques are currently not well-adapted to density estimation, and moreover, density estimators may be slowly converging.

# Assumption-Learn Modeling Estimand

Our estimand is defined as:

$$\beta = \frac{E\{\text{Cov}(A, Y \mid L)\}}{E\{\text{Var}(A \mid L)\}}$$

with efficient influence function:

$$\mathbb{IF}(\beta) = \frac{\{A - E(A \mid L)\}[Y - E(Y \mid L) - \beta\{A - E(A \mid L)\}]}{E[\{A - E(A \mid L)\}^2]}$$

A de-biased estimator which sets its sample average to zero equals:

$$\hat{\beta} = \frac{\sum_{i=1}^n \{A_i - \hat{E}(A_i \mid L_i)\} \{Y_i - \hat{E}(Y_i \mid L_i)\}}{\sum_{i=1}^n \{A_i - \hat{E}(A_i \mid L_i)\}^2}$$

## Question 1 (a) - Relevant Confounders

- **Post-Treatment Variables:** we exclude those variable that may be affected by the treatment, i.e. *smokeN*, *drinkN*, *wgain*, *gestation*.
  - **Mediation Bias:** block part of the causal effect.
  - **Collider Bias:** variable influenced by both the treatment and an unmeasured factor — adjusting for it can open a non-causal path.
- **Are there missing confounders?**
  - For instance, is *income* a missing confounder? Education may be a proxy, but is it enough for the analysis?
  - *Health insurance status*, which may affect both the eligibility for enrolling in the program and the birth weight.



## Question 1 (b) - AL Linear Regression Results

With a dichotomous exposure, our estimand reduces to:

$$\frac{E\{\text{Cov}(A, Y \mid L)\}}{E\{\text{Var}(A \mid L)\}} = \frac{E\{w(L)\beta(L)\}}{E\{w(L)\}}$$

with  $\beta(L) = E(Y \mid A = 1, L) - E(Y \mid A = 0, L)$  and  $w(L) = P(A = 1 \mid L)P(A = 0 \mid L)$ .

**Naïve interpretation** as a mean difference is justified when  $\beta(L) = \beta$ , and **not misleading** otherwise.

## Question 1 (b) - AL Linear Regression Results

If  $\beta(L) = \beta$ , i.e. it is constant, then we would have

$$\begin{aligned}\frac{E\{w(L)\beta(L)\}}{E\{w(L)\}} &= \frac{E\{w(L)\beta\}}{E\{w(L)\}} \\ &= \frac{\beta E\{w(L)\}}{E\{w(L)\}} \\ &= \beta\end{aligned}$$

| Method                    | Estimate | StdError | CI_Lower | CI_Upper |
|---------------------------|----------|----------|----------|----------|
| <chr>                     | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 Linear Model            | -0.00386 | 0.0175   | -0.0381  | 0.0304   |
| 2 Eff. Influence Function | -0.00386 | 0.0175   | -0.0381  | 0.0304   |

- This suggests that participation in the *firststep* program **reduces the probability of low birth weight by 0.39 percentage points**, however...
- Not statistically significant.

## Question 1 (c) - Interpretation when not constant

It is unlikely that  $\beta$  is constant:

- The effect of the program likely depends on **individual characteristics**.
- For instance, **younger or less educated** mothers might gain more from the treatment.
- Some individuals may **already have low risk of low birth weight**, so the program has little room to help.

In this case, we are still computing a weighted average of the main effect. In particular, we are giving **more importance to those strata with an higher variance of the treatment**.

- For instance, if an observation has a propensity score of 0.01, then we are not really interested about his treatment effect since it is very rare that he will receive the treatment.

# Question 1 (d) - Assumption-Lean vs Standard Linear Regression

## Advantages:

- If the model is **misspecified**, we are still guaranteed to capture the association of interest.
- Valid inference is allowed even when using machine learning / model building procedures since we are relying on the estimand's **efficient influence function**.

## Question 1 (e) - Complete Results

|   | Method                  | Estimate | StdError | CI_Lower | CI_Upper |
|---|-------------------------|----------|----------|----------|----------|
|   | <chr>                   | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 | Linear Model            | -0.00386 | 0.0175   | -0.0381  | 0.0304   |
| 2 | Eff. Influence Function | -0.00386 | 0.0175   | -0.0381  | 0.0304   |
| 3 | AIPW                    | 0.0332   | 0.0352   | -0.0358  | 0.102    |
| 4 | TMLE                    | 0.000837 | 0.0130   | -0.0246  | 0.0262   |

`npcausal` package assumes by default a *gaussian* outcome, even though in our case is binary.

## Question 2 (a) - Plug-In Bias

If we directly plug machine learning predictions for the propensity score we will have the so called **plug-in bias**:

- Machine learning algorithms are tailored for predictions (they are regularized in order to minimize the MSE).
- It can be shown that the bias of a plug-in estimator can be computed as  $-1$  over  $N$  times the sample average of the influence functions.

## Question 2 (b) - Weighted Summary Statistics

We can compute the weighted average as:

$$\hat{\theta} = \frac{\sum_{i=1}^n (A_i - \hat{E}[A_i | L_i])^2 X_i}{\sum_{i=1}^n (A_i - \hat{E}[A_i | L_i])^2}$$

(derivation in the appendix...)

The standard error can be estimated as 1 over root  $N$  times the standard deviation the estimand's efficient influence curve:

$$\frac{[P(A = 1 | L)P(A = 0 | L) + \{1 - 2P(A = 1 | L)\}\{A - P(A = 1 | L)\}](X - \theta)}{E\{P(A = 1 | L)P(A = 0 | L)\}}$$

## Question 2 (b) - Weighted Summary Statistics

### Weighted Summary

|           | weighted_mean | weighted_se | weighted_CI_lb | weighted_CI_ub |
|-----------|---------------|-------------|----------------|----------------|
| gender    | 0.525         | 0.015       | 0.496          | 0.555          |
| age       | 26.206        | 0.195       | 25.824         | 26.588         |
| race      | NA            | NA          | NA             | NA             |
| parity    | 0.901         | 0.031       | 0.839          | 0.962          |
| married   | 0.561         | 0.014       | 0.532          | 0.589          |
| smoker    | 0.108         | 0.008       | 0.092          | 0.124          |
| drinker   | 0.008         | 0.003       | 0.003          | 0.013          |
| wpre      | 149.107       | 1.128       | 146.896        | 151.318        |
| education | 12.459        | 0.087       | 12.289         | 12.630         |



## Question 2 (b) - Classic Summary Statistics

### Classic Summary

|           | classic_mean | classic_se | classic_CI_lb | classic_CI_ub |
|-----------|--------------|------------|---------------|---------------|
| gender    | 0.516        | 0.010      | 0.497         | 0.536         |
| age       | 29.304       | 0.120      | 29.069        | 29.539        |
| race      | NA           | NA         | NA            | NA            |
| parity    | 0.819        | 0.021      | 0.778         | 0.860         |
| married   | 0.782        | 0.008      | 0.766         | 0.799         |
| smoker    | 0.070        | 0.005      | 0.060         | 0.080         |
| drinker   | 0.012        | 0.002      | 0.007         | 0.016         |
| wpre      | 146.944      | 0.692      | 145.588       | 148.300       |
| education | 14.072       | 0.053      | 13.969        | 14.175        |

## Question 3 - AL Linear Regression for maternal age

|   | Method                  | Estimate | StdError | CI_Lower  | CI_Upper |
|---|-------------------------|----------|----------|-----------|----------|
|   | <chr>                   | <dbl>    | <dbl>    | <dbl>     | <dbl>    |
| 1 | Linear Model            | 0.00173  | 0.00101  | -0.000258 | 0.00372  |
| 2 | Eff. Influence Function | 0.00173  | 0.00102  | -0.000258 | 0.00372  |

### Interpretation as shift intervention

- A 1-year increase in maternal age is estimated to increase the risk of low birth weight by 0.17% on average, with a 95% confidence interval ranging from a decrease of 0.03% to an increase of 0.37%.
- Note again that the our estimate is not statistically different from 0 (i.e, no effect).

## Question 4 - Results First Steps Program

| Method                    | Estimate | StdError | CI_Lower | CI_Upper |
|---------------------------|----------|----------|----------|----------|
| <chr>                     | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 Linear Model            | 0.0274   | 0.0127   | 0.00249  | 0.0522   |
| 2 Eff. Influence Function | 0.0274   | 0.0127   | 0.00249  | 0.0522   |

- This suggests that participation in the *firststep* program **increases the probability of low birth weight by 2.74 percentage points**, and...
- It is statistically significant at a 5% level.

## Question 4 - Results Maternal Age

| Method                    | Estimate  | StdError | CI_Lower  | CI_Upper  |
|---------------------------|-----------|----------|-----------|-----------|
| <chr>                     | <dbl>     | <dbl>    | <dbl>     | <dbl>     |
| 1 Linear Model            | -0.000276 | 0.000152 | -0.000575 | 0.0000222 |
| 2 Eff. Influence Function | -0.000276 | 0.000152 | -0.000575 | 0.0000223 |

- A 1-year increase in maternal age is estimated to **decrease the risk of low birth weight by 0.028 percentage points** on average, with a 95% confidence interval ranging from a **decrease of 0.058 percentage points** to an **increase of 0.002 percentage points**.
- Note again that the our estimate is not statistically different from 0 (i.e, no effect).

## Question 5 - Risk Ratio

We rely on Assumption-Lean **Loglinear** Regression:

$$\log \{E(Y^a | L)\} = \alpha(L) + \beta a$$

Note that  $\beta$  is defined as:

$$\begin{aligned}\beta &= \log(E[Y^1 | L]) - \log(E[Y^0 | L]) \\ &= \log\left(\frac{E[Y^1 | L]}{E[Y^0 | L]}\right) \\ &= \log(\text{RR})\end{aligned}$$

which leads to

$$\text{RR} = \exp(\beta)$$

## Question 5 - Risk Ratio First Steps Program

|   | Method     | Estimate | StdError | CI_Lower | CI_Upper |
|---|------------|----------|----------|----------|----------|
|   | <chr>      | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 | Risk Ratio | 1.06     | 0.202    | 0.713    | 1.57     |

- Participation in the First Steps program is associated with a **6% higher risk of low birth weight**, compared to non-participation.
- However, this estimate is not statistically significant, as the 95% confidence interval (0.713 to 1.57) includes 1.

## Question 5 - Risk Ratio Age

|   | Method         | Estimate | StdError | CI_Lower | CI_Upper |
|---|----------------|----------|----------|----------|----------|
|   | <chr>          | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 | Risk Ratio Age | 1.00     | 0.00338  | 0.994    | 1.01     |

- A 1-year increase in maternal age does not change the risk of low birth weight in a meaningful way.
- This estimate is (obviously) not statistically significant.

## Question 6 - Odds Ratio

We rely on Assumption-Learn **Logistic** Regression:

$$\text{logit} \{E(Y^a | L)\} = \alpha(L) + \beta a$$

Note that  $\beta$  is defined as:



$$\begin{aligned}\beta &= \text{logit}(E[Y^1 | L]) - \text{logit}(E[Y^0 | L]) \\ &= \log(\text{odds}(Y = 1 | A = 1)) - \log(\text{odds}(Y = 1 | A = 0)) \\ &= \log\left(\frac{\text{odds}(Y = 1 | A = 1)}{\text{odds}(Y = 1 | A = 0)}\right)\end{aligned}$$

which leads to

$$\text{OR} = \exp(\beta)$$



## Question 6 - Odds Ratio First Steps Program

| Method       | Estimate | StdError | CI_Lower | CI_Upper |
|--------------|----------|----------|----------|----------|
| <chr>        | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 Odds Ratio | 1.05     | 0.219    | 0.685    | 1.62     |

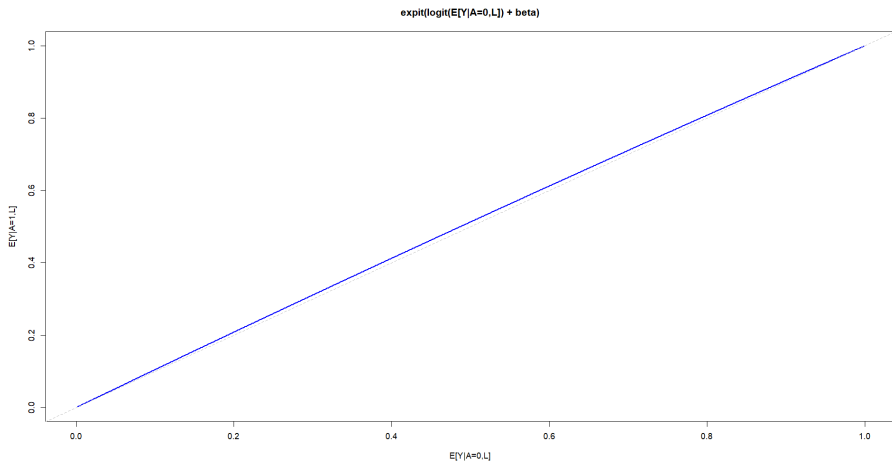
- Participation in the First Steps program is associated with a 5% higher odds of low birth weight, compared to non-participation.
- However, this association is not statistically significant, as the 95% confidence interval (0.685 to 1.62) includes 1.

## Question 6 - Odds Ratio Maternal Age

| Method           | Estimate | StdError | CI_Lower | CI_Upper |
|------------------|----------|----------|----------|----------|
| <chr>            | <dbl>    | <dbl>    | <dbl>    | <dbl>    |
| 1 Odds Ratio Age | 1.58     | 0.471    | 0.626    | 3.97     |

- A 1-year increase in maternal age is associated with a 58% higher odds of low birth weight (Odds Ratio = 1.58).
- The 95% confidence interval ranges from 0.626 to 3.97, suggesting considerable uncertainty in the estimate.
- Because the interval includes 1, the effect is not statistically significant.

## Question 6 (b)



## **APPENDIX**

# General estimand reduces to the weighted average for dichotomous exposures (1)

We want to show that if  $A$  is a dichotomous exposure:

$$\frac{E\{\text{Cov}(A, Y | L)\}}{E\{\text{Var}(A | L)\}} = \frac{E\{w(L)\beta(L)\}}{E\{w(L)\}}$$

Notice that when we define  $w(L) = P(A = 1 | L)P(A = 0 | L)$ , it is already  $\text{Var}(A | L)$  when  $A$  is binary.

Let's focus on the numerator.

$$\text{Cov}(A, Y | L) = E[AY | L] - E[A | L]E[Y | L]$$

## General estimand reduces to the weighted average for dichotomous exposures (2)

- $E[AY \mid L]$

$$\begin{aligned} E[AY \mid L] &= \underbrace{E[AY \mid A = 1, L]}_{\mu_1} \underbrace{P(A = 1 \mid L)}_{p(L)} + \underbrace{E[AY \mid A = 0, L]}_{\mu_0} \underbrace{P(A = 0 \mid L)}_{1-p(L)} \\ &= \mu_1 \cdot p(L) \end{aligned}$$

- $E[A \mid L]$

$$E[A \mid L] = p(L)$$

- $E[Y \mid L]$

$$E[Y \mid L] = \mu_1 \cdot p(L) + \mu_0 \cdot (1 - p(L))$$

## General estimand reduces to the weighted average for dichotomous exposures (3)

Putting all together:

$$\begin{aligned} &= \mu_1 p(L) - p(L) [\mu_1 \cdot p(L) + \mu_0 \cdot (1 - p(L))] \\ &= p(L) [\mu_1 - \mu_1 p(L) - \mu_0 + \mu_0 p(L)] \\ &= p(L) [\beta(L) - p(L) \beta(L)] \\ &= p(L) [\beta(L) (1 - p(L))] \\ &= \beta(L) \underbrace{p(L) (1 - p(L))}_{\text{Var}(A|L)} \end{aligned}$$

$$\text{Cov}(A, Y \mid L) = \beta(L) \text{Var}(A \mid L)$$

$$\beta(L) = \frac{\text{Cov}(A, Y \mid L)}{\text{Var}(A \mid L)}$$

## From the minimizer to the estimand (1)

$$\mathcal{L}(\beta) = E \left[ \int \{E(Y | A + a, L) - E(Y | A, L) - \beta a\}^2 f(A + a | L) da \right].$$

$$\frac{\partial \mathcal{L}(\beta)}{\partial \beta} = \frac{\partial}{\partial \beta} E \left[ \int \{E(Y | A + a, L) - E(Y | A, L) - \beta a\}^2 f(A + a | L) da \right].$$

$$= E \left[ \int -2a \underbrace{\{E(Y | A + a, L) - E(Y | A, L)\}}_{\Delta} - \beta a \} f(A + a | L) da \right]$$

$$= E \left[ -2 \int a \Delta f(A + a | L) da + 2 \int \beta a^2 f(A + a | L) da \right]$$



## From the minimizer to the estimand (2)

Setting the derivative equal to 0 and solving for  $\beta$  we get:

$$\beta = \frac{E \left[ \int a \Delta f(A + a | L) da \right]}{E \left[ \int a^2 f(A + a | L) da \right]}$$

Let  $A' = A + a$ .

**Numerator**

$$\begin{aligned} &= E \left[ \int a (E(Y | A + a, L) - E(Y | A, L)) f(A + a | L) da \right] \\ &= E \left[ \int (A' - A) (E(Y | A', L) - E(Y | A, L)) f(A' | L) dA' \right] \\ &= E \left[ E[(A' - A) (E(Y | A', L) - E(Y | A, L)) | L] \right] \end{aligned}$$

## From the minimizer to the estimand (3)

Setting the derivative equal to 0 and solving for  $\beta$  we get:

**Expand the inner expectation :**

$$= E \left[ (A' - A) E(Y | A', L) | L \right] - E \left[ (A' - A) E(Y | A, L) | L \right].$$

**Term 1:**  $E \left[ (A' - A) E(Y | A', L) | L \right]$

$$\begin{aligned} &= E \left[ A' E(Y | A', L) | L \right] - E \left[ A E(Y | A', L) | L \right] \\ &= E \left[ A' Y | L \right] - E[A | L] E[Y | L] \quad (\text{By LIE}) \end{aligned}$$

**Term 2:**  $E \left[ (A' - A) E(Y | A, L) | L \right]$

$$\begin{aligned} &= -E \left[ A' E(Y | A, L) | L \right] + E \left[ A E(Y | A, L) | L \right] \\ &= -E[A | L] E[Y | L] + E[AY | L] \end{aligned}$$

## From the minimizer to the estimand (4)

Setting the derivative equal to 0 and solving for  $\beta$  we get:

**Combining all together:**

$$= \underbrace{E[A'Y | L] - E[A | L]E[Y | L]}_{\text{Term 1}} - \underbrace{E[A | L]E[Y | L] + E[AY | L]}_{\text{Term 2}}$$

Since  $A'$  and  $A$  are i.i.d. given  $L$  :

$$\begin{aligned} E[A'Y | L] &= E[AY | L] \\ &= 2(E[AY | L] - E[A | L]E[Y | L]) = 2 \text{Cov}(A, Y | L) \\ &= \text{Numerator} = 2E[\text{Cov}(A, Y | L)]. \end{aligned}$$

Similar derivation can be done to the denominator to show that:

$$\text{Denominator} = 2E[\text{Var}(A | L)].$$

## Estimand of Question 2 (1)

$$\theta = \frac{E\{P(A = 1 | L)P(A = 0 | L)X\}}{E\{P(A = 1 | L)P(A = 0 | L)\}}$$

The estimand has efficient influence curve:

$$\text{IF}(\theta) = -\frac{[P(A = 1 | L)P(A = 0 | L) + \{1 - 2P(A = 1 | L)\}\{A - P(A = 1 | L)\}](X - \theta)}{E\{P(A = 1 | L)P(A = 0 | L)\}}$$

Define

- $\phi(A, L) = [P(A = 1 | L)P(A = 0 | L) + \{1 - 2P(A = 1 | L)\}\{A - P(A = 1 | L)\}]$
- $P(A = 1 | L) = p$

## Estimand of Question 2 (2)

Thus we have that

$$\phi(A, L) = p(1 - p) + (1 - 2p)(A - p)$$

$$\text{For } A = 1 \quad = (1 - p)^2$$

$$\text{For } A = 0 \quad = p^2$$

It follows that:

$$\begin{aligned}\phi(A, L) &= A(1 - p)^2 + (1 - A)p^2 \\ &= A(1 + p^2 - 2p) + p^2 - Ap^2 \\ &= A - 2Ap + p^2 \\ &= (A - p)^2\end{aligned}$$

## Estimand of Question 2 (3)

Setting  $E[\text{IF}(\theta)] = 0$  and solving for  $\theta$  we get:

$$\begin{aligned} E[(A - p)^2(X - \theta)] &= 0 \\ E[(A - p)^2X] - \theta E[(A - p)^2] &= 0 \\ \frac{E[(A - p)^2X]}{E[(A - p)^2]} &= \theta \end{aligned}$$

Which can be estimated as:

$$\hat{\theta} = \frac{\sum_{i=1}^n (A_i - \hat{E}[A_i | L_i])^2 X_i}{\sum_{i=1}^n (A_i - \hat{E}[A_i | L_i])^2}$$